Internal Gains Do Not Guarantee External Trustworthiness in Dermatoscopic Classification Under Simulated Label Corruption

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Abstract

We study trustworthy dermatoscopic lesion classification under simulated class-dependent label corruption, budgeted trusted-label repair under simulated oracle supervision, post-hoc calibration, and selective prediction. TrustQueryNet is built around lesion-level group-aware HAM10000 splits, persistent noise and split manifests, multi-seed evaluation, and external validation on the official ISIC 2019 test set. Using a corrected ConvNeXt-Tiny training protocol with explicit best-checkpoint evaluation, the main repair configuration achieved 0.8350 ± 0.0059 internal calibrated accuracy, 0.7152 ± 0.0216 macro-F1, and 0.9382 ± 0.0133 macro-AUROC across five seeds. However, strong baselines remained highly competitive: no-repair training reached 0.7145 ± 0.0130 internal macro-F1, random repair reached 0.7105 ± 0.0239, and a clean-label upper bound reached 0.7330 ± 0.0288. External validation revealed a substantial performance drop for all approaches. On ISIC 2019, the repair model reached 0.5692 ± 0.0145 calibrated accuracy and 0.4427 ± 0.0117 macro-F1, compared with 0.5630 ± 0.0078 and 0.4288 ± 0.0123 for no repair, and 0.5591 ± 0.0203 and 0.4311 ± 0.0328 for random repair. Internal temperature scaling did not resolve external calibration fragility, and generalized cross-entropy collapsed under the chosen corruption regime. An overlap audit found zero exact duplicate images between HAM10000 and the ISIC 2019 external slice. The resulting evidence supports a restrained but clinically relevant conclusion: modest internal gains from trusted-label repair do not imply broad external trustworthiness, and repair strategies must be judged against strong baselines, including random repair.

1 Introduction

Dermatoscopic lesion classification is a useful test bed for trustworthy machine learning because it combines severe class imbalance, multiple images per lesion, dataset shift across collections, and imperfect labels. The HAM10000 dataset remains one of the most common dermatoscopic benchmarks for this problem domain [1]. At the same time, clinically realistic external validation is known to be substantially harder than internal benchmarking, as highlighted by the ISIC 2019 challenge validation study [2]. In this setting, a model can look strong in-distribution while remaining poorly calibrated or externally brittle.

TrustQueryNet was developed to evaluate these issues directly rather than treating them as secondary diagnostics. The project combines four ideas in one reproducible pipeline:

- lesion-level, group-aware splitting for HAM10000;
- explicit class-dependent noisy-label simulation;
- budgeted trusted-label repair under simulated oracle supervision;
- calibration and selective prediction analysis under both internal and external evaluation.

This paper is not positioned as a new method paper. Its value is as a rigorous, externally validated trustworthy-ML study asking a practical question: when labels are noisy and trusted

correction is budget-limited, do modest internal gains from repair survive stronger baselines and external shift?

2 Related Work

HAM10000 provides a widely used dermatoscopic benchmark with multiple images per lesion and a clinically relevant class taxonomy [1]. For external robustness, the ISIC 2019 challenge validation study is especially relevant because it demonstrates how skin lesion classifiers degrade under more realistic distribution shift and out-of-training-category conditions [2].

For the image backbone, we use ConvNeXt-Tiny, following the ConvNeXt architecture family [3]. Calibration is handled with post-hoc temperature scaling, a standard and strong baseline for neural probability calibration [4]. Selective prediction is summarized through risk-coverage analysis and AURC-style outputs, grounded in the selective classification literature [5]. As a noisy-label comparison, we include generalized cross-entropy (GCE), a well-known robust-loss baseline [6]. The main corrected recipe also uses light label smoothing [7].

3 Methods

3.1 Data and split protocol

HAM10000 is the primary development dataset. TrustQueryNet performs lesion-level grouping so that images from the same lesion do not leak across train, validation, and test partitions. Split manifests are written to disk and reused across runs. Clean labels and observed labels are tracked separately, alongside trust and repair state.

The external dataset is the official ISIC 2019 test set [2]. Labels are mapped into the HAM10000-style seven-class space:

- MEL → mel
- NV → nv
- BCC → bcc
- AK → akiec
- SCC → akiec
- BKL → bkl
- DF → df
- VASC → vasc
- UNK filtered

3.2 Noise and repair protocol

Observed training labels are corrupted by a fixed class-dependent transition matrix. A fraction of samples is marked as initially trusted, and subsequent repair rounds replace $y_{observed}$ with y_{clean} for selected samples. This is a retrospective simulation of limited expert correction, not a prospective clinician-in-the-loop study. The most accurate description is therefore *budgeted trusted-label repair under simulated oracle supervision.*

3.3 Models and evaluation

The final corrected study uses ConvNeXt-Tiny [3] with:

- 12 epochs;
- cross-entropy with 0.05 label smoothing [7] for the main repair and comparison baselines;
- explicit best-checkpoint selection by validation macro-F1;
- temperature scaling [4];
- dense selective metrics including AURC [5];
- multi-seed aggregation.

We report calibrated accuracy, macro-F1, macro-AUROC, expected calibration error (ECE), AURC, coverage at confidence 0.5, and risk at confidence 0.5.

4 Experimental Protocol

The corrected internal evidence suite includes:

- **Repair**: budgeted entropy-based trusted-label repair;
- **No repair**: identical training recipe without repair rounds;
- **Random repair**: matched repair budget with random sample selection;
- **Clean upper**: clean-label upper bound without simulated corruption;
- **GCE no repair**: robust-loss baseline under the same noisy setup [6].

The main external evidence suite evaluates the three most relevant operational variants: repair, no repair, and random repair. An image-overlap audit between HAM10000 and ISIC 2019 reports exact-hash matches and perceptual-hash near-duplicate candidates.

5 Results

5.1 Internal main comparison

Setting	Accuracy	Macro-F1	ECE	Macro-AUROC	AURC	Coverage@0.5	Risk@0.5
Repair	0.8350 ± 0.0059	0.7152 ± 0.0216	0.0445 ± 0.0090	0.9382 ± 0.0133	0.0728 ± 0.0186	0.9525 ± 0.0074	0.1428 ± 0.0054
No repair	0.8321 ± 0.0071	0.7145 ± 0.0130	0.0356 ± 0.0075	0.9460 ± 0.0128	0.0622 ± 0.0093	0.9480 ± 0.0066	0.1449 ± 0.0078
Random repair	0.8319 ± 0.0051	0.7105 ± 0.0239	0.0356 ± 0.0071	0.9521 ± 0.0128	0.0588 ± 0.0145	0.9376 ± 0.0100	0.1402 ± 0.0107

Table 1: Corrected internal main comparison on HAM10000.

The internal comparison does not support a "repair wins everywhere" story. Repair produces a small point-performance gain in accuracy and macro-F1, but no-repair and random-repair remain highly competitive, and both outperform repair on some trust metrics such as ECE, AUROC, and AURC.

5.2 Noisy-label anchors

The clean upper bound shows moderate remaining headroom, while the generalized cross-entropy baseline fails dramatically under the chosen noisy-label regime. This negative result is important: robust-loss substitution is not automatically beneficial in this setting.

Setting	Accuracy	Macro-F1	ECE	Macro-AUROC	AURC	Coverage@0.5	Risk@0.5
Repair	0.8350 ± 0.0059	0.7152 ± 0.0216	0.0445 ± 0.0090	0.9382 ± 0.0133	0.0728 ± 0.0186	0.9525 ± 0.0074	0.1428 ± 0.0054
Clean upper	0.8521 ± 0.0055	0.7330 ± 0.0288	0.0389 ± 0.0176	0.9583 ± 0.0076	0.0502 ± 0.0095	0.9618 ± 0.0107	0.1306 ± 0.0089
GCE no repair	0.6774 ± 0.0012	0.1224 ± 0.0122	0.0268 ± 0.0381	0.5782 ± 0.1279	0.2399 ± 0.1059	0.9639 ± 0.0626	0.3044 ± 0.0327

Table 2: Noisy-label anchors and baseline comparisons.

5.3 External validation

Setting	Accuracy	Macro-F1	ECE	Macro-AUROC	AURC	Coverage@0.5	Risk@0.5
Repair external	0.5692 ± 0.0145	0.4427 ± 0.0117	0.2000 ± 0.0253	0.8125 ± 0.0180	0.2804 ± 0.0299	0.8526 ± 0.0230	0.3805 ± 0.0228
No repair external	0.5630 ± 0.0078	0.4288 ± 0.0123	0.2016 ± 0.0123	0.8168 ± 0.0154	0.2749 ± 0.0183	0.8498 ± 0.0264	0.3838 ± 0.0073
Random repair external	0.5591 ± 0.0203	0.4311 ± 0.0328	0.2149 ± 0.0395	0.8114 ± 0.0337	0.2905 ± 0.0669	0.8654 ± 0.0383	0.3964 ± 0.0269

Table 3: External validation on the official ISIC 2019 test set.

All methods degrade sharply on ISIC 2019 relative to internal HAM10000 evaluation. Repair remains modestly ahead of the two comparison baselines on external accuracy and macro-F1, but it does not dominate all trust metrics. The overall lesson is not that repair solves external robustness, but that even modest internal gains must be interpreted cautiously under shift.

5.4 Overlap audit

The overlap audit found 10,015 HAM10000 samples, 6,191 mapped ISIC 2019 external samples, zero exact duplicate images, and 14,185 perceptual-hash candidate pairs at dHash ≤ 4. The zero exact matches are the key integrity result. The perceptual-hash list is a coarse candidate screen for manual or secondary review and should not be interpreted as confirmed leakage.

6 Discussion

The final evidence supports a restrained but useful conclusion. Budgeted trusted-label repair is not a universally dominant intervention in this setting. It yields only a modest advantage over no-repair and random-repair baselines in point performance, while other trust metrics remain mixed. This is exactly why the stronger baseline suite matters: without random repair and no repair, it would have been easy to overstate the value of repair.

The external findings are the most important part of the paper. Internal HAM10000 performance is strong for all competitive variants, but all of them degrade substantially on ISIC 2019. That drop is not a side note; it is the central trustworthy-ML result. Similarly, post-hoc calibration learned in-distribution does not remove external confidence fragility, echoing the broader cautionary lessons of realistic external evaluation in dermoscopy [2].

The paper is therefore strongest when framed as an externally validated experimental study rather than a method paper. Its contribution is evidence: which interventions remain competitive, which fail, and how internal trust signals do or do not transfer externally.

7 Limitations

- The repair protocol is simulated oracle supervision, not prospective expert annotation.
- External validation is limited to one external dataset.
- Perceptual-hash overlap candidates were not exhaustively adjudicated manually.
- Repair does not dominate all trust metrics, so claims must remain modest.
- No clinician reader study or deployment evaluation is included.

- The work does not introduce a new learning algorithm.

8 Conclusion

TrustQueryNet is now a credible applied research package for trustworthy dermatoscopic classification under noisy supervision. The final evidence shows that internal gains do not guarantee external trustworthiness, that trusted-label repair must be judged against strong baselines including random repair, and that robust-loss substitution can fail outright in this regime. These are publishable applied findings when presented honestly and with external validation.

References

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